Other Services Inventory

Last updated by | Roy Leonardo Siroy | Aug 26, 2025 at 9:32 PM GMT+3

AWS Services Inventory

This document provides a inventory of AWS services used in the NGPS Global project, excluding Lambda functions

Services Overview

Service Category	AWS Service	Resource Type	Usage Description	Configuration Files
Compute	API Gateway	REST API	Main API endpoints for the application	`gatewayResponse/` , `gatewayValidator/`
Storage	S3	Buckets	File storage for exports, analysis data, and backups	`s3/buckets.yml`
Storage	DynamoDB	Tables	Primary database for application data	`dynamoDb/tables.yml` , `dynamoDb/tables-extended.yml`
Storage	DynamoDB DAX	Caching Cluster	In-memory caching for DynamoDB	`dynamoDb/dax.yml`
Caching	ElastiCache Redis	Clusters	Redis caching for performance optimization	`redis/elasticCacheClusters.yml`
Messaging	SQS	Queues	Message queuing for asynchronous processing	`sqs/queues.yml`
Streaming	Kinesis	Data Streams	Real-time data streaming for AEP exports	`kinesis/aepExportDataStream.yml` , `kinesis/partTranPrizeCatTablesWithStreams.`
Security	IAM	Roles & Policies	Identity and access management	`iam/iamGroupsAndPolicies.yml` , `iam/iamLambdaRole.yml`
Security	Cognito	Identity Pools	Authentication and authorization	`cognito/SelfServiceCognito.yml` , `cognito/SelfServiceCognitoRoles.yml`

Security	WAF v2	Web ACLs	Web application firewall	`waf/webACLs.yml`
Networking	VPC	Virtual Private Cloud	Network isolation and security	`vpc/vpc.yml`
Networking	CloudFront	Distribution	Content delivery network	`cloudFront/cloudFrontDist.yml`
Monitoring	CloudWatch	Alarms & Dashboards	Monitoring and alerting	`cloudWatch/alarms.yml` , `cloudWatch/dashboards.yml` , `cloudWatch/metrics.yml`
Monitoring	CloudWatch	Circuit Breaker	Circuit breaker pattern implementation	`cloudWatch/circuitBreaker.yml`
Monitoring	X-Ray	Sampling Rules	Distributed tracing	`xRay/xray.yml`
Configuration	SSM	Parameter Store	Configuration management	`ssmParameters/ssmParameters.yml`
Data Processing	CloudWatch Logs	Log Groups	Log archiving and processing	`cloudWatchLogArchiver/`
Backup	AWS Backup	Backup Plans	Data backup and retention	`backups/retentionStrategy.yml`
Audit	CloudTrail	Audit Logs	API call logging and auditing	`oneTimeResourcesCreation/globalAuditLog.yml`

Service Dependencies and Integrations

Primary Service	Dependent Services	Integration Type	Purpose
DynamoDB	Kinesis Data Streams	Native Streams	Real-time data export for AEP
DynamoDB	DAX	Caching Layer	Performance optimization
API Gateway	WAF v2	Security Layer	Request filtering and rate limiting
Lambda Functions	VPC	Network Access	Secure resource access
ElastiCache Redis	VPC	Network Isolation	Secure caching within private network
S3	CloudWatch	Monitoring	Bucket access and lifecycle monitoring
All Services	IAM	Authorization	Role-based access control
All Services	CloudWatch	Monitoring	Comprehensive observability
All Services	X-Ray	Tracing	Distributed request tracing

Environment-Specific Configurations

Feature	Environment Variable	Purpose	Conditional Deployment
Data Export	`createDataExporter`	Controls DynamoDB streams for export	✅
Data Archiving	`archiveExpiredConfigData`	Controls archiving functionality	✅
Redis Caching	`useRedis`	Enables/disables Redis clusters	✅
AEP Export V1	`enableParticiaptionAEPStream`	Legacy AEP export stream	✅
AEP Export V2	`enableAepExportV2`	Enhanced AEP export functionality	✅
OU Segmentation	`enableOuSegmentationForAep`	Multi-region AEP processing	✅
DynamoDB Backups	`enableDynamoDbBackups`	Point-in-time recovery	✅
CloudWatch Alarms	`createAlarms`	Monitoring and alerting	✅
WAF Protection	`createWAF`	Web application firewall	✅
Leaderboard	`useLeaderboard`	Leaderboard functionality with Redis	✅

Regional Deployment Strategy

Service	Multi-Region Support	Region Strategy	Notes
Kinesis Data Streams	✅	EMEA, NOAM, APAC specific streams	Regional data processing
S3	✅	Cross-region replication available	Configurable per bucket
DynamoDB	✅	Global tables support	Cross-region replication
CloudFront	✅	Global edge locations	Automatic global distribution
ElastiCache Redis	❌	Single region per cluster	Multiple clusters for HA
VPC	❌	Single region	Regional network isolation

Cost Optimization Features

Service	Cost Optimization	Implementation
S3	Lifecycle Policies	Glacier transition after 7 days
DynamoDB	On-Demand Pricing	Auto-scaling based on usage
Kinesis	On-Demand Mode	Pay per request model
ElastiCache	Right-sizing	Environment-specific instance types
CloudWatch	Log Retention	Stage-specific retention policies

Security Compliance

Security Control	Implementation	Services Involved
Encryption at Rest	Default encryption	S3, DynamoDB, ElastiCache
Encryption in Transit	TLS/SSL	All HTTP/API communications
Network Isolation	VPC with private subnets	All compute resources
Access Control	IAM roles and policies	All AWS services
Audit Logging	CloudTrail	All API calls
WAF Protection	Rate limiting, IP filtering	API Gateway

Operational Excellence

Aspect	Implementation	Tools Used
Monitoring	Comprehensive dashboards	CloudWatch, X-Ray
Alerting	Multi-level alarm system	CloudWatch Alarms, SNS
Circuit Breaking	Fault tolerance patterns	CloudWatch, Custom metrics
Backup & Recovery	Automated backup strategies	AWS Backup, DynamoDB PITR
Configuration Management	Environment-based configs	SSM Parameter Store